Al-Isra university
Faculty of Engineering
Electrical Engineering Department
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Second Exam
Signal Processing and Systems

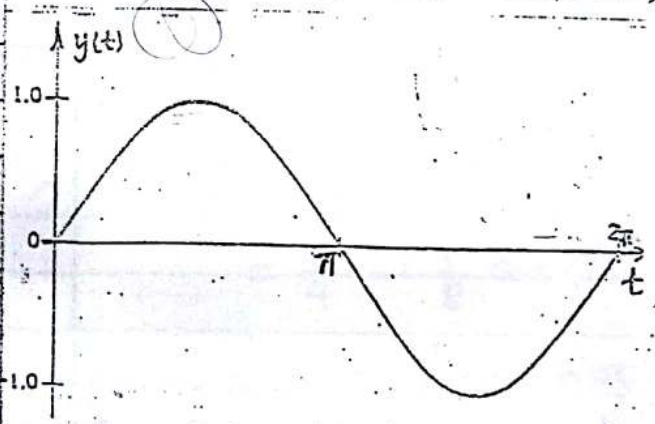
17

1. **[7 marks]** The complex exponential Fourier Series of a periodic signal with period T is given by $x(t) = \sum (4\text{sinc}(n/4))\exp(-j20\,\pi nt)$ for all n. Find:
   a. The average value. b. The fundamental frequency.
   c. The amplitude and the phase of the third harmonic.
   d. X_n, a_n, and b_n for $n < 4$.
   e. The average power contained in the first three harmonics.

2. **[8 marks]** Consider the signal signal $x(t) = \cos^2(100\pi t)\sin^2(150\pi t)$. Find:
   a. The complex coefficients X_n.
   b. The fundamental frequency.
   c. The average power of the signal.

3. **[10 marks]** Using Table 4-1 and Table 4-2, find:
   A. The Inverse Fourier Transform of:
      a. $X(f) = \exp(j4f)\Lambda(0.5f)$.
      b. $Y(f) = 4\cos(10\,\pi f) / |4 + j2\pi f|^2$.
   B. The Fourier Transform of the signals shown in Fig. 1.

4. **[Bonus]** Consider a fixed linear system with amplitude response $A + A.\Pi[f/200]$ and phase response $-0.01\pi f$ for $|f| < 50$, 0.5π for $f < -50$, and -0.5π for $f > 50$. Find the output and determine the type of distortion for the following inputs:
   a. $y(t) = \cos(120\pi t) + \cos(220\pi t)$. b. $w(t) = \cos(80\pi t) + 4\cos(160\pi t)$.



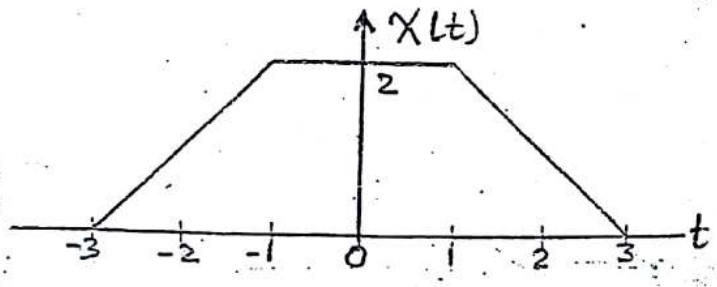


Fig. 1

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1 $x(t) = \sum_{n=-\infty}^{\infty} 4\,\text{sinc}\left(\frac{n}{4}\right) e^{-j20\pi nt}$ $\quad 0 \le t \le T.$

$X_n = 4\,\text{sinc}\left(\frac{n}{4}\right)$

1 a. The average value $= X_0 = 4$.

1 b. $20\pi = 2\pi f_0 \rightarrow f_0 = 10$ Hz.

1.5 c. $X_3 = 4\,\text{sinc}\left(\frac{3}{4}\right) = 4\,\frac{\sin(3\pi/4)}{3\pi/4} = 1.2$

$|X_3| = 1.2, \quad \theta_3 = 0.$

2.5 d. $X_n = \frac{1}{2} a_n$ for all n since X_n is real.

$X_0 = 4, \quad X_1 = 4\,\text{sinc}\left(\frac{1}{4}\right) = 4\,\frac{\sin(\frac{1}{4}\pi)}{(1/4)\pi} = 3.6.$

$X_2 = 4\,\text{sinc}\left(\frac{1}{2}\right) = 2.55, \quad X_3 = 4\,\text{sinc}\left(\frac{3}{4}\right) = 1.2$

$X_{-n} = X_n, \quad a_0 = 4, \; a_1 = 7.2, \; a_2 = 5.1, \; a_3 = 2.4.$

1 e. $P = X_0^2 + 2\sum_{n=1}^{3} |X_n|^2 = 57.8$

2 $x(t) = \cos^2(100\pi t)\sin^2(150\pi t) = \frac{1}{4}[1+\cos(200\pi t)][1-\cos(300\pi t)]$

2 $= \frac{1}{4} - \frac{1}{8}\cos(100\pi t) + \frac{1}{4}\cos(200\pi t) - \frac{1}{4}\cos(300\pi t)$

$- \frac{1}{8}\cos(500\pi t)$

3 a. $a_0 = \frac{1}{4}, \; a_1 = -\frac{1}{8}, \; a_2 = \frac{1}{4}, \; a_3 = -\frac{1}{4}, \; a_5 = -\frac{1}{8}.$

$X_0 = \frac{1}{4}, \quad X_1 = -\frac{1}{16}, \quad X_2 = \frac{1}{8}, \quad X_3 = -\frac{1}{8}$

$X_5 = -\frac{1}{16}.$

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2 b. The fundamental frequency = 50.

1

2 c. $P = X_0^2 + 2\sum_{n=1}^{\infty} |X_n|^2 = (\frac{1}{4})^2 + 2[(\frac{1}{16})^2 + (\frac{1}{8})^2 + (\frac{1}{8})^2 + (\frac{1}{16})^2]$

$= \frac{9}{64}$.

3 A. a. $X(f) = e^{j4f}\,\Lambda(0.5f) = \frac{1}{1.2} e^{-j2\pi f(\frac{2}{\pi})} \cdot 2\Lambda(\frac{f}{2})$.

$x(t) = 2\,\text{sinc}^2(2(t+\frac{2}{\pi}))$

b. $Y(f) = \frac{4\cos(2\pi f(5))}{16 + (2\pi f)^2} = \frac{1}{2}\cos(2\pi(5)f)\,\frac{2(4)}{(4)^2 + (2\pi f)^2}$.

$y(t) = \frac{1}{4}[\delta(f+5) + \delta(f-5)] * e^{-4|t|}$

$= \frac{1}{4}[e^{-4|t+5|} + e^{-4|t-5|}]$.

B. $y(t) = \Pi(\frac{t-\pi}{2\pi})\sin(t)$

$Y(f) = 2\pi e^{-j2\pi^2 f}\,\text{sinc}(2\pi f) * \frac{1}{2j}[\delta(f - \frac{1}{2\pi}) + \delta(f + \frac{1}{2\pi})]$

$= -j\pi\, e^{-j2\pi^2(f-\frac{1}{2\pi})}\,\text{sinc}(2\pi(f-\frac{1}{2\pi})) +$

$j\pi\, e^{-j2\pi^2(f+\frac{1}{2\pi})}\,\text{sinc}(2\pi(f+\frac{1}{2\pi}))$.

$x(t) = 3\Lambda(\frac{t}{3}) - \Lambda(t)$.

$X(f) = 9\,\text{sinc}^2(3f) - \text{sinc}^2(f)$.

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